

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Eksamen

Kopiereg voorbehou

Examination

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Module EBN111
Elektrisiteit en Elektronika
18 Junie 2012

Module EBN111
Electricity and Electronics
18 June 2012



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

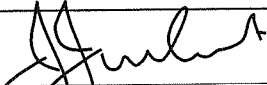
Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	90	Volpunte: <i>Full marks:</i>	90
Duur van vraestel: <i>Duration of paper:</i>	180 minute <i>180 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		26	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Polêre vorm: $A\angle\theta$; Reghoekige vorm: $a + jb$
Polar form: $A\angle\theta$; Rectangular form:
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert		
	2 JD le Roux		
Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding. <i>Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.</i>			
Groephoof: <i>Group Head:</i>	Prof J Joubert	Handtekening: <i>Signature:</i>	

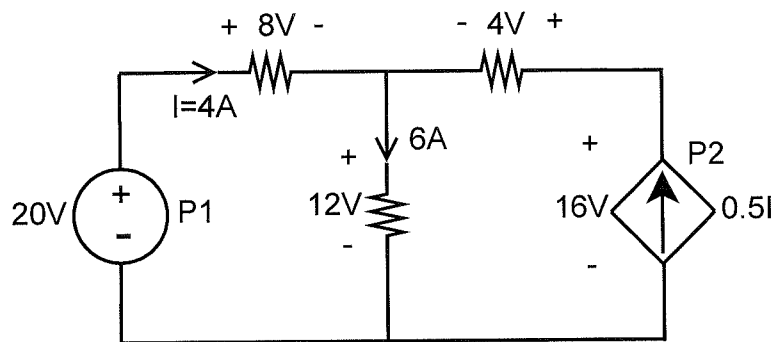
VRAAG/QUESTION 1 [25]

Vraag 1.1[6]

- (a) Bepaal die stroom wat by $t = 2\text{ s}$ deur 'n element vloei as die ladingsvloei gegee word deur $q(t) = (3t^2 + 4t - 2)\text{ C}$.
- (b) 'n Plasma-skerm TV gebruik 339 W drywing, en 'n LCD-skerm TV gebruik 213 W. As elke TV vir 5 ure per dag vir 365 dae gebruik word, wat sal die kostebesparing wees vir die LCD TV oor die jaar as die koste 75 c/kWh is.
- (c) Bepaal die drywing ge-assosieer met die spannings-bron (P1) en die stroombron (P2) in die stroombaan.

Question 1.1[6]

- (a) Determine the current flowing through an element at $t = 2\text{ s}$ if the charge flow is given by $q(t) = (3t^2 + 4t - 2)\text{ C}$.
- (b) A plasma screen TV draws 339 W of power, and a LCD screen TV draws 213 W. If each set was used for 5 hours a day for 365 days, what would the cost savings be for the LCD unit over the year if the cost is 75 c/kWh.
- (c) Find the power associated with the voltage source (P1) and the current source (P2) in the circuit.



Vraag 1.2[4]

- (a) Bepaal I en V_{ab} in die stroombaan van Fig 1.2(a).
(b) Bepaal i en v in die stroombaan van Fig 1.2(b).

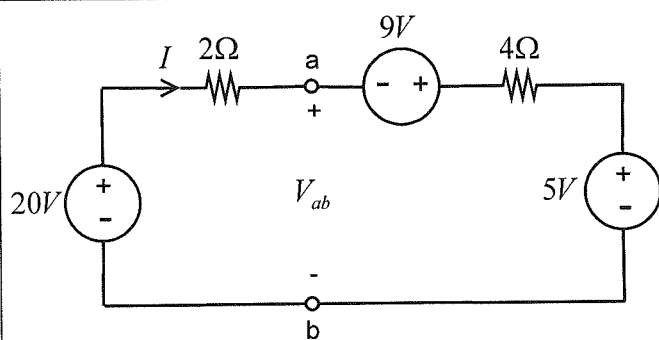


Fig 1.2(a)

Question 1.2[4]

- (a) Determine I and V_{ab} in the circuit of Fig 1.2(a).
(b) Determine i and v in the circuit of Fig 1.2(b).

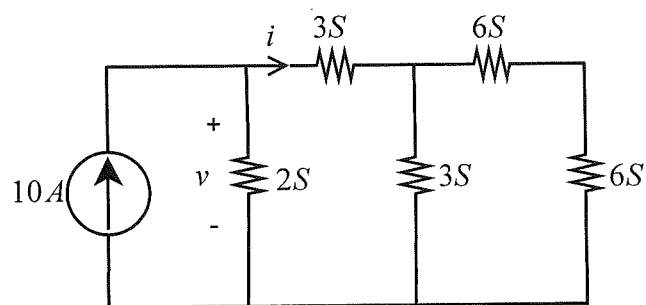


Fig 1.2(b)

Vraag 1.3[4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$aV_1 + bV_2 = 50$$

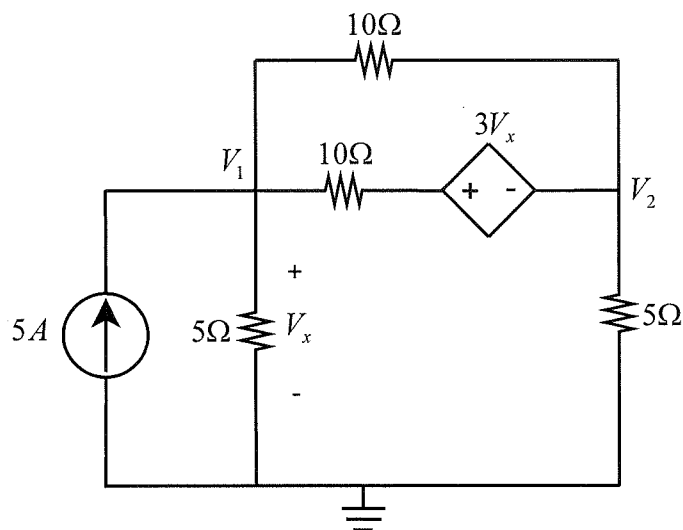
$$cV_1 + dV_2 = 0$$

Question 1.3[4]

Use nodal analysis to formulate two equations of the following form:

$$aV_1 + bV_2 = 50$$

$$cV_1 + dV_2 = 0$$



Vraag 1.4[4]

Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel:

$$6i_1 + ei_3 = f$$

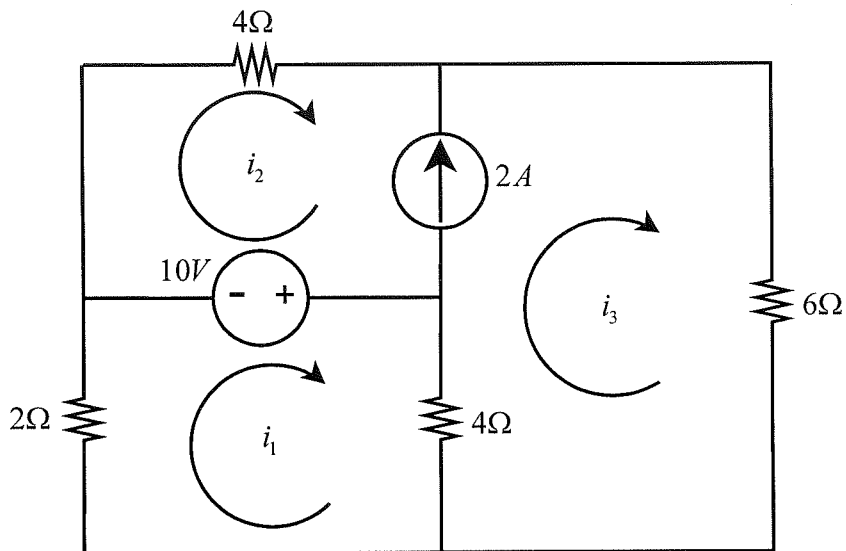
$$-4i_1 + gi_3 = h$$

Question 1.4[4]

Use mesh analysis to formulate two equations of the following form:

$$6i_1 + ei_3 = f$$

$$-4i_1 + gi_3 = h$$

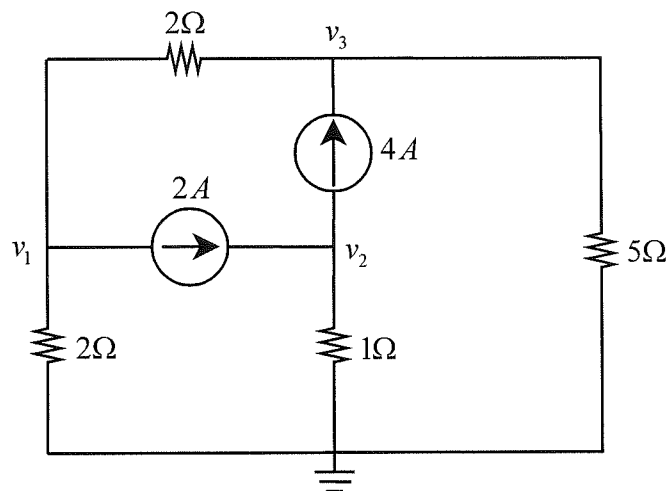


Vraag 1.5[3]

Skryf die node-spannings matriks van die stroombaan neer deur **inspeksie** (jy mag nie KCL gebruik nie, en skryf antwoorde as reële getalle, en nie breuke nie).

Question 1.5[3]

Write the node-voltage matrix for the circuit by **inspection** (you may not use KCL, and write the answers as real numbers, and not fractions).

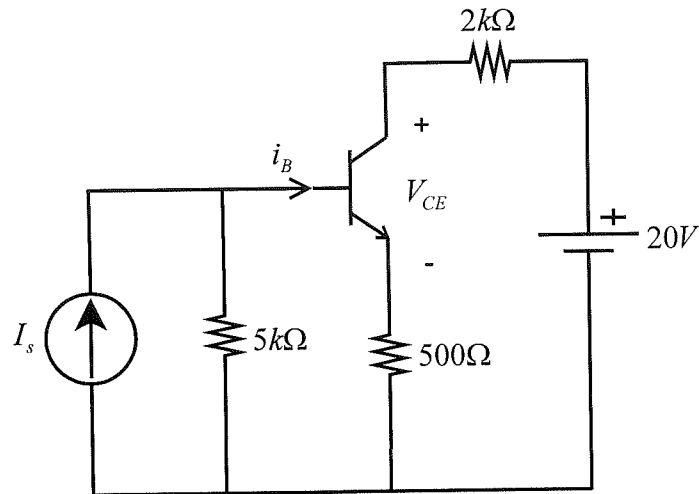


Vraag 1.6[4]

- (a) Bereken I_B (in μA) as $I_s = 1 \text{ mA}$, $\beta = 100$ en $V_{BE} = 0.7\text{V}$.
(b) Bereken V_{CE} as $I_B = 50 \mu\text{A}$, $\beta = 100$ en $V_{BE} = 0.7\text{V}$.

Question 1.6[4]

- (a) Calculate I_B (in μA) if $I_s = 1 \text{ mA}$, $\beta = 100$ and $V_{BE} = 0.7\text{V}$.
(b) Calculate V_{CE} if $I_B = 50 \mu\text{A}$, $\beta = 100$ and $V_{BE} = 0.7\text{V}$.



VRAAG/QUESTION 2 [20]

Vraag 2.1[4]

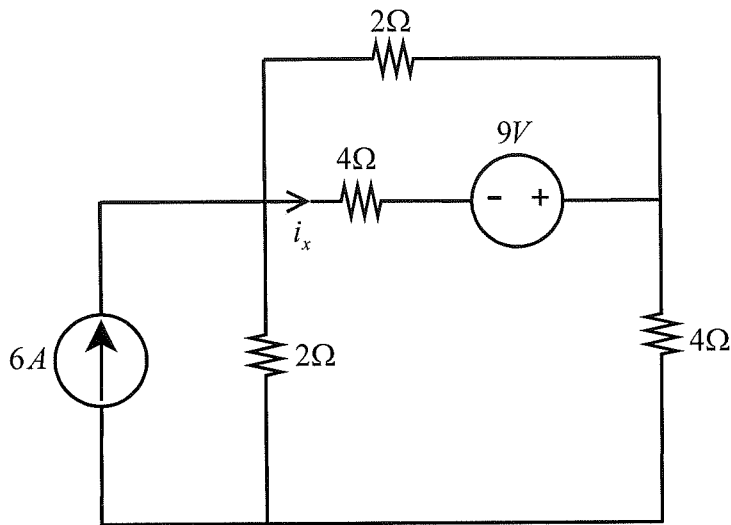
Gebruik superposisie:

- (a) Bereken i_x agv die 9V spanningsbron.
- (b) Bereken i_x agv die 6A stroombron.

Question 2.1[4]

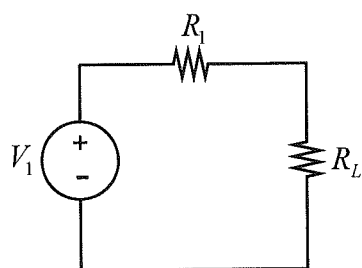
Use superposition:

- (a) Determine i_x due to the 9V voltage source.
- (b) Determine i_x due to the 6A current source.



Vraag 2.2[4]

Gebruik brontransformasies om die stroombaan in Fig 2.2 te vereenvoudig na:

**Question 2.2[4]**

Use source transformations to simplify the circuit in Fig 2.2 to:

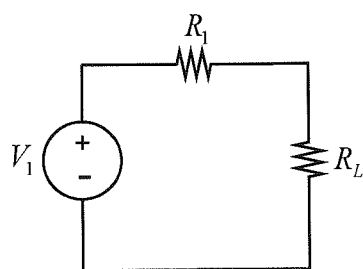
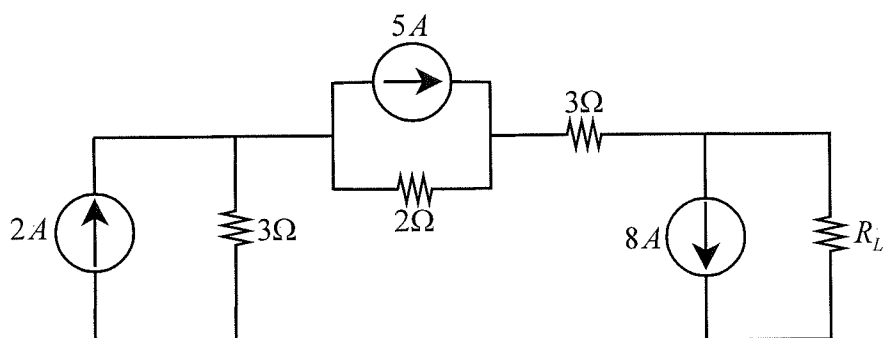


Fig 2.2

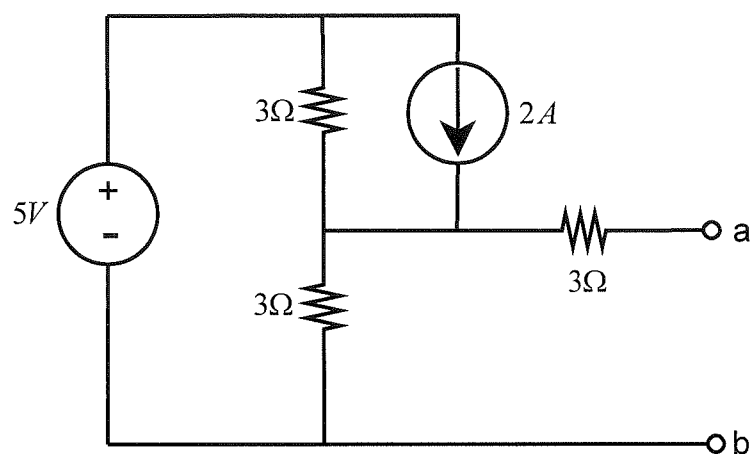


Vraag 2.3[4]

Bepaal die Thevenin ekwivalent tov terminale a-b.

Question 2.3[4]

Determine the Thevenin equivalent wrt terminals a-b.

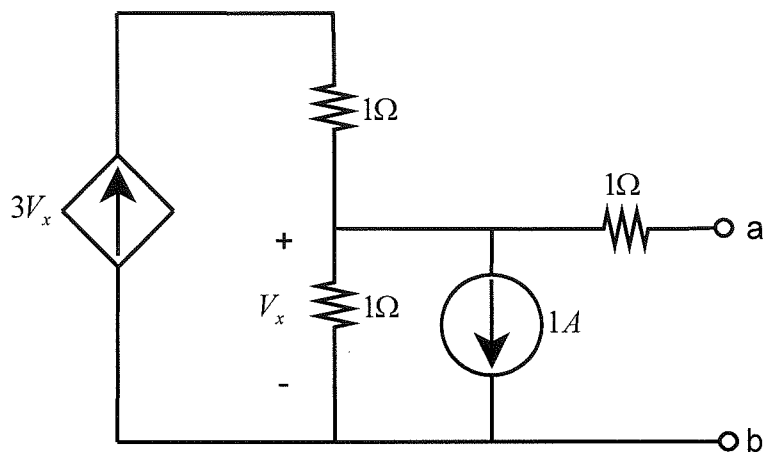


Vraag 2.4 [4]

Bepaal die Norton ekwivalent tov terminale a-b.

Question 2.4 [4]

Determine the Norton equivalent wrt terminals a-b.

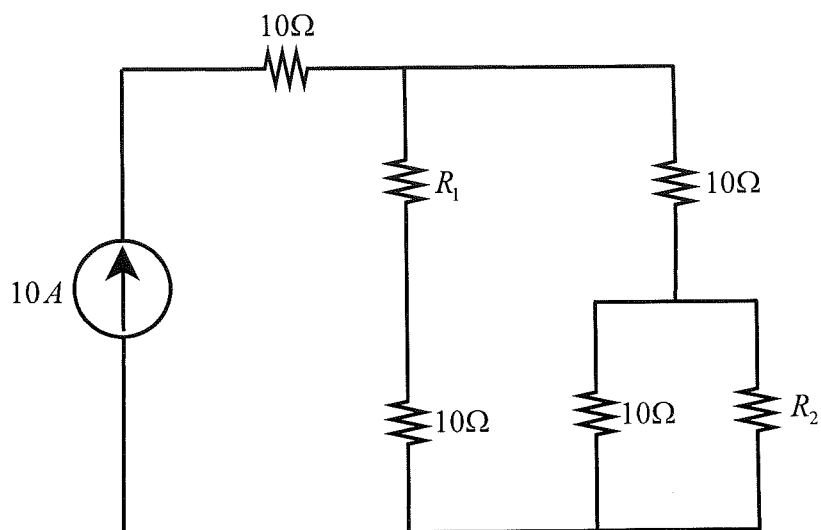


Vraag 2.5[4]

- (a) Aanvaar $R_1 = 10\ \Omega$ en $R_2 = R_L$ - bepaal die waarde van R_L vir maksimum drywingsoordrag na R_L .
- (b) Aanvaar $R_1 = R_L$ en $R_2 = 10\ \Omega$ - bepaal die waarde van R_L vir maksimum drywingsoordrag na R_L .

Question 2.5[4]

- (a) Assume $R_1 = 10\ \Omega$ en $R_2 = R_L$ - find the value of R_L for maximum power transfer to R_L .
- (b) Assume $R_1 = R_L$ en $R_2 = 10\ \Omega$ - find the value of R_L for maximum power transfer to R_L .



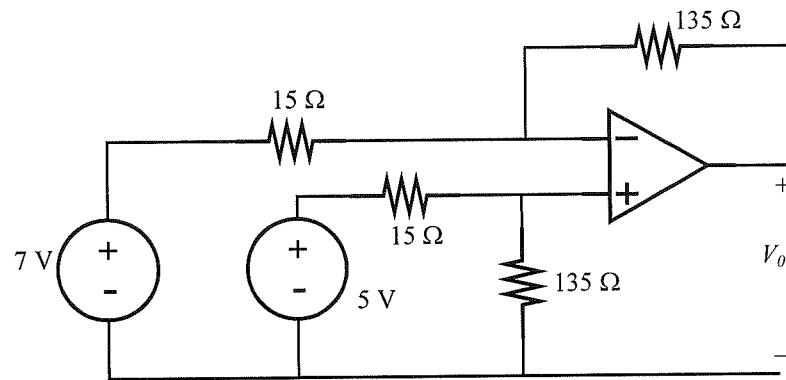
VRAAG/QUESTION 3 [20]

Vraag 3.1 [3]

Bepaal V_0 .

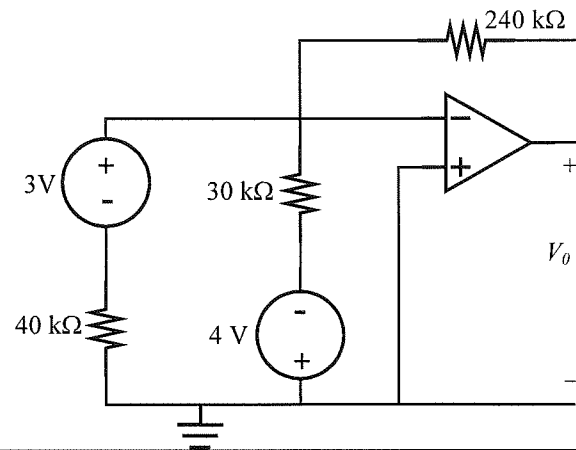
Question 3.1 [3]

Determine V_0 .



Vraag 3.2 [3]
Bepaal V_o .

Question 3.2 [3]
Determine V_o .



Vraag 3.3 [4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan:

$$aV_1 + bV_0 = 12$$

$$V_1 = cV_0$$

(Punte toedeling: a - [1]; b - [1]; c - [2])

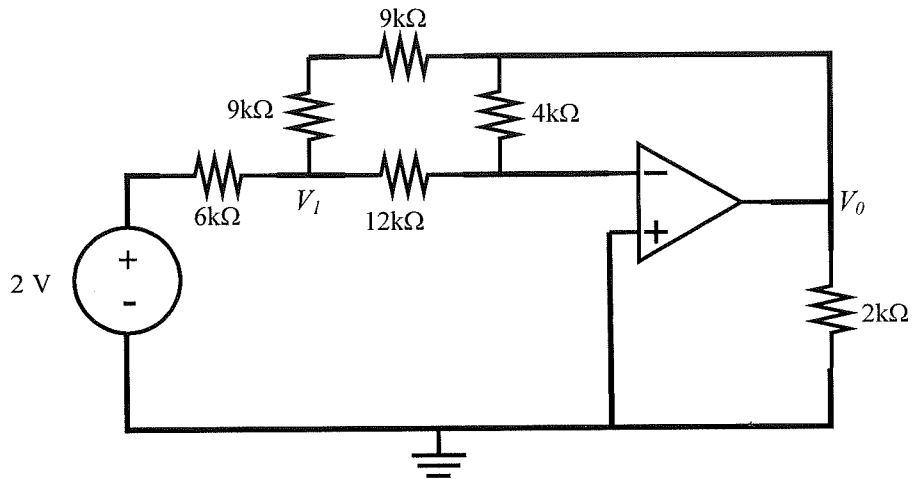
Question 3.3 [4]

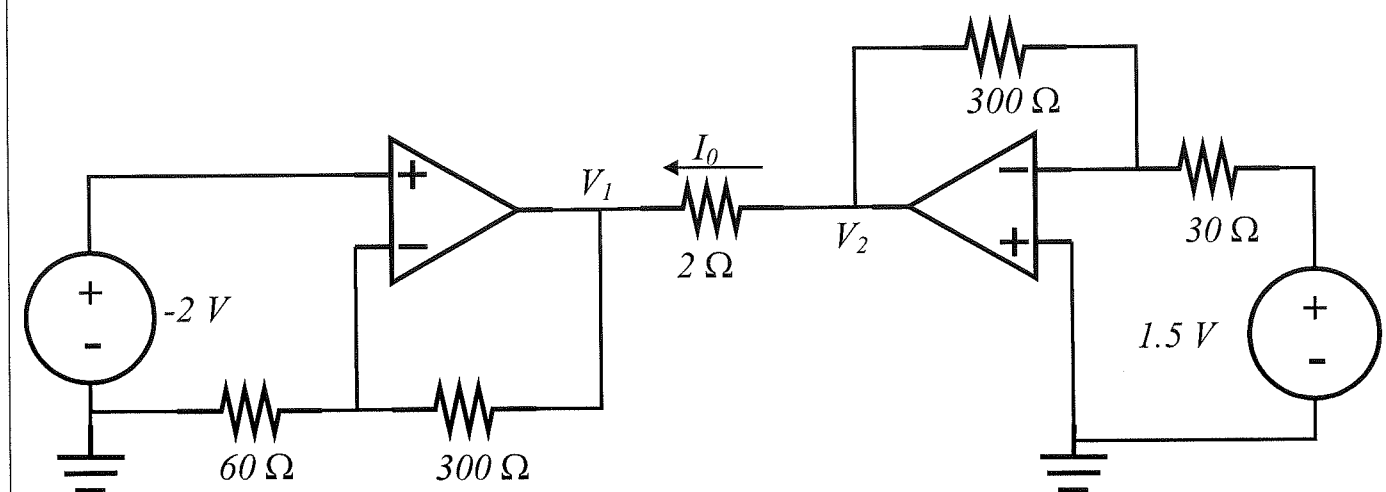
Use node analysis to set up two equations of the following form for the circuit:

$$aV_1 + bV_0 = 12$$

$$V_1 = cV_0$$

(Mark allocation: a - [1]; b - [1]; c - [2])



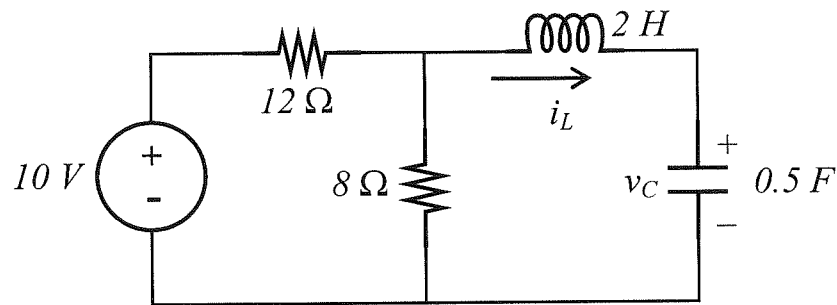
Vraag 3.4 [5]Vir die kaskade kringbaan, bepaal V_1 , V_2 en I_0 .(Punte toedeling: V_1 - [2]; V_2 - [2]; I_0 - [1])**Question 3.4 [5]**For the cascade circuit, determine V_1 , V_2 and I_0 .(Mark allocation: V_1 - [2]; V_2 - [2]; I_0 - [1])

Vraag 3.5 [3]

Vir DC-toestande, bepaal die energie gestoor deur die kapasitor en induktor. (Sluit asseblief die eenheid vir energie by die antwoord in.)

Question 3.5 [3]

For DC conditions, determine the energy stored by the inductor and capacitor. (Please include the unit for energy in the answer.)

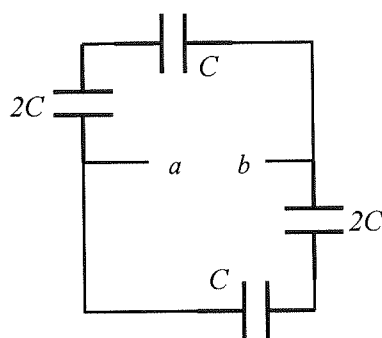


Vraag 3.6 [4]

a) Bepaal die ekwivalente kapasitansie tussen terminale a en b in terme van C . [2]

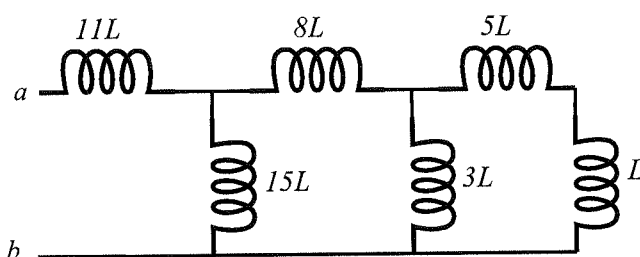
Question 3.6 [4]

a) Determine the equivalent capacitance between terminals a and b in terms of C . [2]



b) Bepaal die ekwivalente induktansie tussen terminale a en b in terme van L . [2]

b) Determine the equivalent inductance between terminals a and b in terms of L . [2]



VRAAG/QUESTION 4 [25]

Vraag 4.1 [2]

a) Is V_1 voorlopend of nalopend tot V_2 ? Met hoeveel grade? [1]

b) Is V_1 voorlopend of nalopend tot V_3 ? Met hoeveel grade? [1]

(Skryf die antwoord as volg:

V_1 is voorlopend/nalopend tot V_2 met φ°)

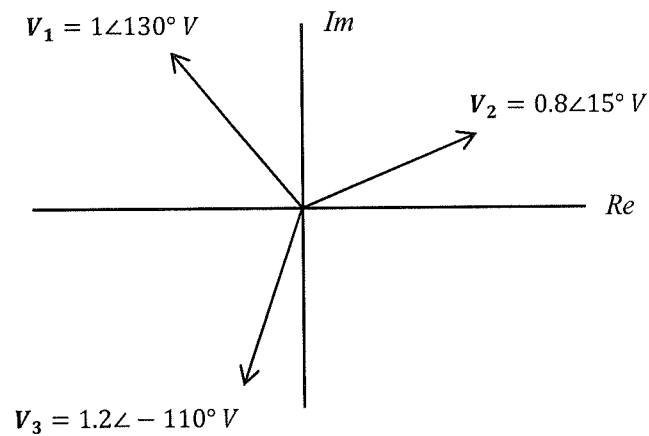
Question 4.1 [2]

a) Is V_1 leading or lagging V_2 ? With how many degrees? [1]

b) Is V_1 leading or lagging V_3 ? With how many degrees? [1]

(Write the answer as follows:

V_1 is leading/lagging V_2 with φ°)



Vraag 4.2 [2]

Gebruik fasors om die volgende as 'n enkele cosinus uitdrukking $[A \cos(\omega t + \sigma), A > 0, |\sigma| \leq 180^\circ]$ te skryf:

$$v(t) = -10 \cos(\omega t - 120^\circ) - 5 \sin(\omega t + 30^\circ) \text{ V}$$

Question 4.2 [2]

Use phasors to write the following as a single cosine expression $[A \cos(\omega t + \sigma), A > 0, |\sigma| \leq 180^\circ]$:

$$v(t) = -10 \cos(\omega t - 120^\circ) - 5 \sin(\omega t + 30^\circ) \text{ V}$$

Vraag 4.3 [2]

Evalueer die volgende en skryf die antwoord in **polêre** vorm:

$$\left(\frac{-12 - 9j}{5e^{j36.8699^\circ}} + (3 + 3j)^* \right)^{-1}$$

Question 4.3 [2]

Evaluate and write the final answer in **polar** form:

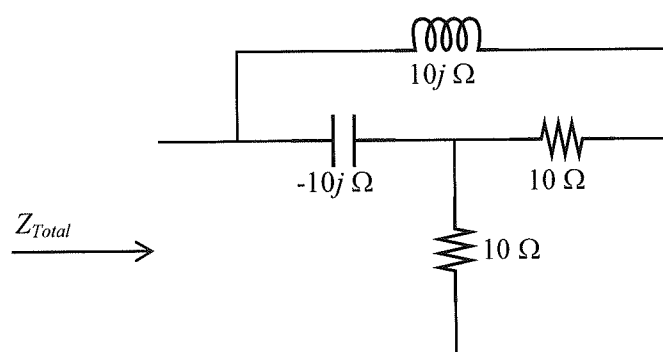
$$\left(\frac{-12 - 9j}{5e^{j36.8699^\circ}} + (3 + 3j)^* \right)^{-1}$$

Vraag 4.4 [2]

Bepaal Z_{Total} . Skryf die finale antwoord in **reghoekige** vorm.

Question 4.4 [2]

Determine Z_{Total} . Write the final answer in **rectangular** form.



Vraag 4.5 [5]

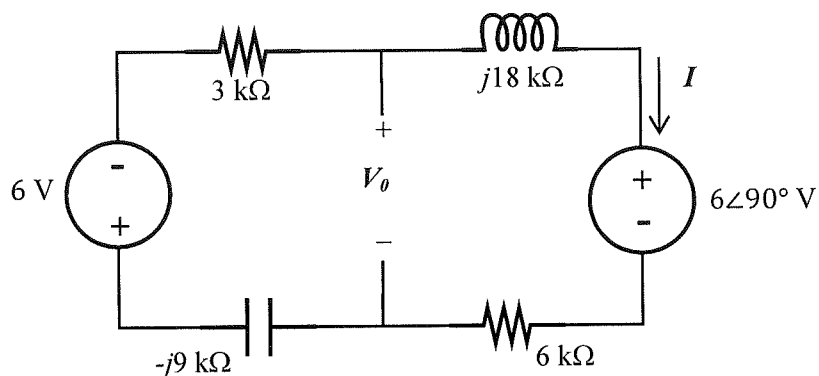
Vir onderstaande kringbaan, bepaal:

- a) ω in rad/s indien $L = 100$ mH. [1]
- b) die kapasitansie C indien $\omega = 50$ rad/s. [1]
- c) V_θ en I in **reghoekige** vorm. [3]

Question 4.5 [5]

For the circuit below, determine:

- a) ω in rad/s if $L = 100$ mH. [1]
- b) the capacitance C if $\omega = 50$ rad/s. [1]
- c) V_θ and I in **rectangular** form. [3]



Vraag 4.6 [4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan:

$$aV_1 + bV_2 = 1\angle 50^\circ$$

$$cV_1 - dV_2 = j20$$

Die koëffisiënte a tot d moet in **reghoekige** vorm geskryf word.

(Punte toedeling: a & b - [2]; c & d [2])

Question 4.6 [4]

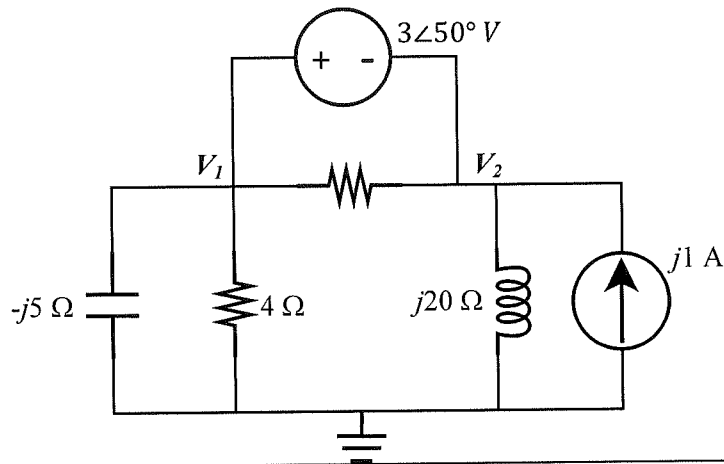
Use node analysis to set up two equations of the following form for the circuit:

$$aV_1 + bV_2 = 1\angle 50^\circ$$

$$cV_1 - dV_2 = j20$$

The coefficients a to d must be written in **rectangular** form.

(Mark allocation: a & b - [2]; c & d [2])



Vraag 4.7 [4]

Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan:

$$aI_1 + j25I_2 = b$$

$$j25I_1 + cI_2 = d$$

Die koeffisiente a tot d moet in **reghoekige** vorm geskryf word.

(Punte toedeling: a & b - [2]; c & d [2])

Question 4.7 [4]

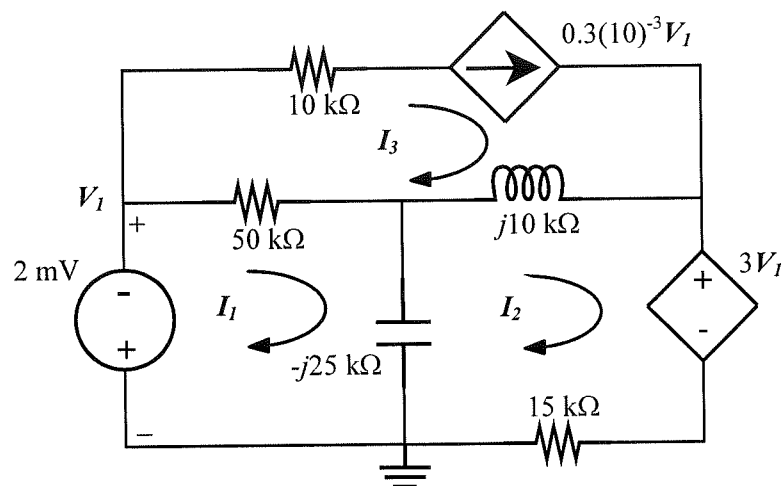
Use mesh analysis to set up two equations of the following form for the circuit:

$$aI_1 + j25I_2 = b$$

$$j25I_1 + cI_2 = d$$

The coefficients a to d must be written in **rectangular** form.

(Mark allocation: a & b - [2]; c & d [2])



Vraag 4.8 [4]

Bepaal die Thevenin ekwivalent tussen terminale a en b vir onderstaande kringbaan. Skryf die antwoord in **polêre** vorm.

(Wenk: Gebruik 'n $1\angle 0^\circ$ A toetsbron of 'n $1\angle 0^\circ$ V spanningstoetsbron om Z_{TH} te bereken.)

Question 4.8 [4]

Determine the Thevenin equivalent between terminals a and b of the circuit shown below. Write the answer in **polar** form.

(Hint: Use a $1\angle 0^\circ$ A test current source or a $1\angle 0^\circ$ V test voltage source to determine Z_{TH} .)

